

Supplemental Materials for Continual Adaptation for Efficient Machine Communication

Anonymous EMNLP submission

Appendix A: derivation of incremental KL

We denote the distribution over a sequence of T tokens by $p(w_{1:T}) = p(w_1, w_2, \dots, w_T)$. We are interested in the KL divergence between two such distributions, $KL(p(w_{1:T}) \parallel q(w_{1:T}))$. We show that our approximation over possible captions is the best incremental estimator of this intractable objective. First, note that the KL divergence factors in the following way.

Lemma 1.

$$\begin{aligned} & KL(p(w_{1:2}) \parallel q(w_{1:2})) \\ &= KL(p(w_1) \parallel q(w_1)) \\ &\quad + \mathbb{E}_{p(w_1)} KL(p(w_2|w_1) \parallel q(w_2|w_1)) \end{aligned}$$

Proof.

$$\begin{aligned} & KL(p(w_{1:2}) \parallel q(w_{1:2})) \\ &= \sum_{w_1} \sum_{w_2} p(w_{1:2}) \log \frac{p(w_{1:2})}{q(w_{1:2})} \\ &= \sum_{w_1} \sum_{w_2} p(w_{1:2}) \log \frac{p(w_1)}{q(w_1)} \\ &\quad + \sum_{w_1} \sum_{w_2} p(w_{1:2}) \log \frac{p(w_2|w_1)}{q(w_2|w_1)} \\ &= \sum_{w_1} \log \frac{p(w_1)}{q(w_1)} \sum_{w_2} p(w_{1:2}) \\ &\quad + \sum_{w_1} p(w_1) \sum_{w_2} p(w_2|w_1) \log \frac{p(w_2|w_1)}{q(w_2|w_1)} \\ &= KL(p(w_1) \parallel q(w_1)) \\ &\quad + \mathbb{E}_{p(w_1)} KL(p(w_2|w_1) \parallel q(w_2|w_1)) \end{aligned}$$

Now, let w_1^* be the token at which $p(w_1)$ takes its maximum value. Then w_1^* is the best single-sample approximation of the expectation:

$$\begin{aligned} & \mathbb{E}_{p(w_1)} KL(p(w_2|w_1) \parallel q(w_2|w_1)) \\ & \approx KL(p(w_2|w_1^*) \parallel q(w_2|w_1^*)) \end{aligned}$$

If we assume that $p(w_{1:T})$ is Markov (as in a recurrent model) then it follows from repeatedly applying the lemma that

$$\begin{aligned} & KL(p(w_{1:T}) \parallel q(w_{1:T})) \\ &= \sum_{i=1}^T KL(p(w_i|w_1^*, \dots, w_{i-1}^*) \parallel q(w_i|w_1^*, \dots, w_{i-1}^*)) \\ &= \sum_{i=1}^T KL(p(w_i|w_{i-1}^*) \parallel q(w_i|w_{i-1}^*)) \end{aligned}$$

recovering our objective.

Appendix B: Parameter settings

For both the speaker task and listener task, we used a learning rate of 0.0005, took 6 gradient steps after each trial, and used a batch size of 8 when sampling utterances from the augmented set of sub-phrases. At each gradient step, we sampled 50 objects from the full domain $\mathcal{O}$ of COCO to approximate our regularization term. We set the coefficients weighting each term in our loss function as follows: 1.0 (utterance loss), 0.1 (contrastive loss), 0.5 (KL regularization), 0.3 (local rehearsal).

Appendix C: Regression details

To formally test increases in efficiency reported for baseline pairs of humans in Sec. 4.1 (see Fig. 3 and S2), we conducted a mixed-effects regression predicting utterance length. We included a fixed effect for context type (i.e. ‘simple’ vs. ‘challenging’) as well as orthogonalized linear and quadratic effects of repetition number, and each of their interactions with context type. We also included random intercepts accounting for variability in initial utterance length at the pair- and image-level. To test increases in the adaptive listener model’s accuracy in Sec. 4.2, we conducted a mixed-effects logistic regression on trial-level responses (i.e. ‘correct’